

Hidden Regimes

A neural autoencoder and a hidden Markov model, chained together, learn four recurring market regimes from 19 years of equity, volatility, credit, rates, and commodity data — with no P&L objective and no human-labeled examples.

SAMPLE

2007-01-01 → 2026-08-18

TRAIN / HOLDOUT SPLIT

through 2021-12-31 / 2022-2026

UNIVERSE

12 tickers — equity, vol, credit, rates, FX, commodities

PREPARED

2026-08-19

CURRENT REGIME ● Normal as of 2026-08-18	CONFIDENCE 98.5% posterior probability	STATES SELECTED 4 of 2–7 swept	GFC CAPTURED 94% days in Crisis state	COVID CAPTURED 92% days in Crisis state
REFIT AGREEMENT 81.6–83.7% across 4 vintages				

01 Executive Summary

This is a standalone research model, not a trading strategy — it has no backtest, no Sharpe ratio, and nothing in this report should be read as a signal to size a position. Its job is narrower: does a fully unsupervised machine learning pipeline, shown 19 years of macro/vol/credit/rate data and never told what a "crisis" is, discover regime structure that lines up with regimes a human would recognize?

It does, with real nuance. Two AI components — a neural-network autoencoder for representation learning, and a hidden Markov model for regime discovery (Section 4 explains both in detail) — jointly settle on four persistent states. The rare, extreme one captures 94% of the 2008 financial

crisis and 92% of the COVID crash by trading days, while correctly placing the grinding, slower-moving 2022 rate-hike bear market in a *different* state entirely — the model separates sharp panics from grinding bear markets without ever being told the difference. It is moderately, not perfectly, stable under refitting: about 5 in 6 historical days keep their label as new data arrives.

As of 2026-08-18, the model reads the market as being in its "Normal" state with 98.5% confidence — an elevated-but-not-alarming regime, distinct from both the calm low-vol state and the rare Crisis state.

02 What This Model Does

Markets don't behave the same way at all times — volatility clusters, correlations shift, calm periods and panics have different statistical character. A "regime" is a period during which the market's behavior is internally consistent, followed by a transition to a different consistent behavior. This model doesn't assume regimes exist and then look for confirmation; it asks an unsupervised learning pipeline to find the number and shape of regimes that best explain 19 years of market data, with no labels, no crisis dates, and no target variable supplied at any point.

The deliverable is a daily regime label and a probability distribution over all four regimes, going back to 2008 and updated live through the most recent trading day — useful as a macro-context layer (e.g., for conditioning a separate trading strategy's risk appetite, though that integration is not built here), not as a strategy in itself.

03 Data & Universe

A deliberately diverse macro/vol/credit/rates/commodity/breadth set, not just equity prices — the wider and more genuinely different the raw inputs, the more there is for the autoencoder to actually discover.

TICKER	ROLE
SPY	Broad equity
RSP	Equal-weight equity – vs. SPY, a concentration/breadth proxy
IWM	Small caps – risk appetite proxy
QQQ	Growth/tech – style rotation proxy
^VIX	Implied volatility
HYG	High-yield credit
LQD	Investment-grade credit – vs. HYG, a credit-stress proxy
TLT	Long treasuries – rates/duration proxy
GLD	Gold – safe-haven flows
UUP	Dollar index – USD flows
USO	Oil – commodity vol proxy
^TNX	10-year Treasury yield level

Sourced from Yahoo Finance's public chart API, daily bars, 2007-01-01 onward — chosen specifically to reach back before the 2008 financial crisis, the sample's single most important stress-test event. 16 causal features are derived from these 12 series before anything AI-related happens: log returns, cross-asset return spreads (e.g. HYG – LQD as a credit-stress measure), rolling realized volatility, and rolling z-scores of level variables like the VIX and the 10-year yield. Every feature at date t uses only data dated on or before t — no feature ever looks forward.

04 Where AI Is Used

Two distinct machine learning components, chained together, do all of the actual regime discovery. No large language model, generative AI, or synthetic data is used anywhere in this pipeline — both components below are classical/deep *unsupervised* learning, trained directly on real market data with no human-provided labels.

Neural network autoencoder

A small feedforward neural network (PyTorch), architecture `16 → 12 → 3 → 12 → 16` with ReLU activations, trained to compress the 16 raw daily features down to a 3-dimensional latent embedding and then reconstruct the original 16 from just those 3 numbers. It is trained by gradient descent (Adam optimizer, 300 epochs) to minimize reconstruction error (MSE) between the original and reconstructed feature vectors.

This is the AI system deciding, on its own, what 3-dimensional summary of "the state of the market today" best preserves the information spread across 16 noisy, correlated inputs — instead of a person hand-picking 2–3 "important" indicators. It never sees regime labels; its only objective is faithful reconstruction.

↓ latent embedding (3 numbers per day) ↓

Gaussian hidden Markov model

A probabilistic sequence model (`hmmlearn`) fit on the autoencoder's 3-dimensional output. It learns, purely from the statistical structure of the sequence: how many distinct hidden states best explain the data, each state's characteristic mean and covariance in latent space, and a transition matrix — the probability of the market staying in its current regime vs. switching to another, day to day. This is what actually produces "regime" as a concept: a state is inferred to exist because the model finds it statistically useful for explaining runs of similar days, never because any date was tagged as belonging to one.

Applied to new data, the fitted HMM outputs both a most-likely regime label per day and a full posterior probability across all states — the 98.5% confidence figure on the cover of this report is exactly that model output for the most recent trading day.

Why chain two AI systems instead of one

Each does the job it's suited for. The autoencoder is good at compressing a wide, messy, correlated raw feature set into a compact summary — but has no notion of persistence or sequence. The HMM is good at discovering persistent, sequential structure — but works best on a small number of well-behaved dimensions, not 16 noisy raw features directly. Feeding the HMM the autoencoder's 3-dimensional learned summary, rather than either the full 16 raw features or a small hand-picked subset, was a deliberate choice to let representation learning and regime discovery each do their own job.

Training discipline — no lookahead in either AI stage

Both the autoencoder's weights and the HMM's parameters are fit using **only** data dated on or before 2021-12-31. Both are then frozen and applied forward, unmodified, to 2022 through the present — so nothing in the 2022–2026 holdout period, including the live read on the cover of this report, could have influenced either model's learned parameters. Autoencoder holdout reconstruction loss (0.70) sits only mildly above train loss (0.56) — the healthy case; a holdout loss far above train would flag overfitting to the training window.

05 Model Selection — Why 4 States, Not More

The HMM was swept across 2–7 candidate state counts. BIC (a standard model-selection statistic that penalizes complexity) improves monotonically all the way to 7 — on that metric alone, more states always look better. But BIC doesn't know what a "regime" should mean; it will happily reward the model for carving out a state that lasts 1.7 days and flickers back and forth, exactly as if it had found a genuine new multi-week regime.

STATES	TRAIN LOG-LIK / OBS	BIC	SHORTEST STATE'S AVG DURATION
2	−3.311	23,523	14.6 days
3	−2.866	20,494	10.8 days
4 (selected)	−2.624	18,923	6.9 days
5	−2.529	18,400	1.7 days ✕
6	−2.508	18,412	1.7 days ✕
7	−2.418	17,959	1.6 days ✕

The rule actually used: pick the *largest* state count, among those tried, where *every* state still has a plausible average duration of at least 5 trading days. That lands on 4 — the point immediately before BIC's further "improvements" start being explained by noise-fitting rather than genuine new structure, visible directly in the collapsing minimum-duration column above.

06 Validation — Three Checks, No Sharpe Ratio

With no P&L objective, "is this model good" needs a different answer than a backtest. Three checks, escalating from statistical to economic to robustness:

1. Held-out log-likelihood

Already visible in the table above — the chosen 4-state model's fit doesn't degrade from train to holdout, evidence it generalizes rather than having memorized the training window.

2. Historical crisis alignment

The model was never told any of these dates. If its rare, extreme-return state reliably lands on real, independently-known stress events, that is genuine external validation.

EVENT	DAYS	CALM BULL	CRISIS	NORMAL	TURBULENT BULL
2008 GFC	146	—	94%	—	6%
2011 Eurozone crisis	85	—	75%	15%	9%
2015–16 China/oil selloff	145	9%	10%	28%	53%
Dec 2018 selloff	63	11%	3%	51%	35%
COVID crash	52	—	92%	8%	—
2022 rate-hike bear market	209	3%	1%	34%	62%

THE INTERESTING FINDING

The Crisis state dominates the two sharp, liquidity/credit-driven panics (2008, COVID) but is nearly absent from the 2022 bear market and the milder 2015-16 and Dec-2018 selloffs — those land mostly in "Normal" or "Turbulent Bull" instead. The model separates fast panics from grinding, slower-moving drawdowns on its own, purely from the statistical shape of the data — a distinction it was never given any information about.

3. Refit stability

The full pipeline (autoencoder + HMM) is retrained from scratch at 4 progressively later cutoffs, and regime labels for a shared 2009–2014 window (inside every vintage's own training data) are compared. HMM state indices are arbitrary between independent fits, so states are matched across vintages by nearest centroid on (mean return, volatility) before comparing labels directly.

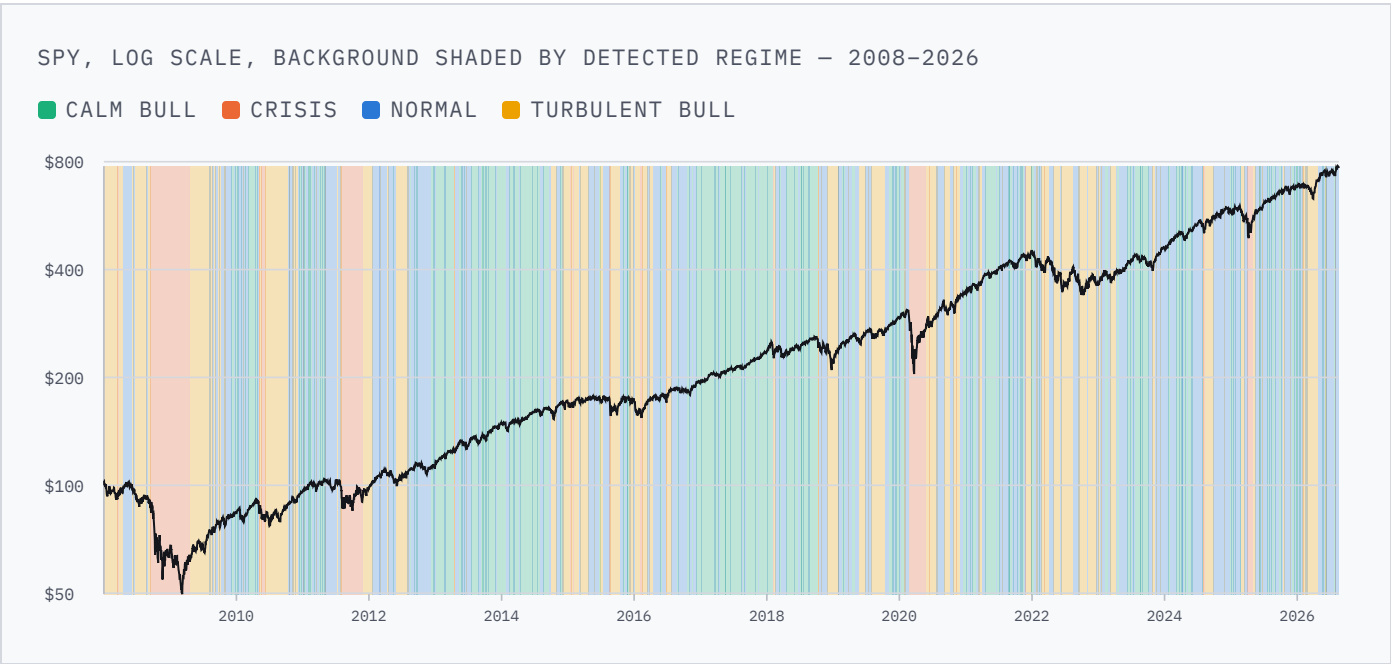
VINTAGE (TRAIN_END)	STATES SELECTED	AGREEMENT WITH FINAL VINTAGE, 2009–2014
2015-12-31	4	81.6%
2018-12-31	4	82.7%
2021-12-31	4	83.7%
2025-12-31 (reference)	5	100.0% (self)

Moderately stable, not perfect — roughly 1 day in 6 over the shared window gets relabeled as more data arrives, concentrated in transition/boundary periods rather than spread evenly. Worth flagging on its own terms: the automatic state-count rule (Section 5) is not perfectly stable either — the 2025-12-31 vintage's independent sweep selected **5** states, not 4, given nearly the full sample to work with. The production model in this report intentionally freezes its own selection at the 2021-12-31 vintage rather than re-selecting live, so this doesn't affect the report's current regime read, but it does mean the automatic rule itself could plausibly favor a different state count if refit today.

07 Regime Characterization

REGIME	DAYS	SHARE	SPY ANN. RETURN	SPY ANN. VOL	AVG DURATION
● Calm Bull	1,241	26.5%	+15.2%	8.3%	8.4 days
● Crisis	387	8.3%	−53.1%	48.9%	20.7 days
● Normal	1,821	38.9%	+16.4%	13.2%	6.9 days
● Turbulent Bull	1,238	26.4%	+25.5%	20.1%	12.0 days

SPY return/vol stats are descriptive characterization of each state after the fact, not model inputs — the HMM never sees SPY's own return, only the 3-dimensional latent embedding built from all 16 features.



Transition matrix

Row = current regime, column = probability of tomorrow's regime.

FROM \ TO	CALM BULL	CRISIS	NORMAL	TURBULENT BULL
● Calm Bull	88.1%	0.3%	11.4%	0.2%
● Crisis	0.0%	95.2%	0.0%	4.8%
● Normal	9.9%	0.3%	85.4%	4.5%
● Turbulent Bull	0.0%	1.3%	7.1%	91.6%

Crisis is both the rarest state to enter and, once entered, the second-most persistent (95.2% chance of staying day to day, ~21-day average episode) — consistent with panics being sharp to arrive but slow to fully resolve. Direct Calm-Bull-to-Crisis and Crisis-to-Calm-Bull transitions are essentially zero in 19 years of data; the model has learned that the market passes through an intermediate state on the way into or out of a genuine panic, not that it treats state-switching as unconstrained.

08 Current Read

REGIME	POSTERIOR PROBABILITY, 2026-08-18
● Calm Bull	0.44%
● Crisis	0.003%
● Normal	98.48%
● Turbulent Bull	1.08%

The model reads current conditions as "Normal" with high confidence — historically the most common state (38.9% of the full sample), characterized by positive drift and moderate (13.2% annualized) volatility, distinct from both the calmer, lower-vol Calm Bull state and the two higher-vol bull/crisis states. This is a description of the model's own inferred latent structure, not a market forecast or a trading signal.

09 Known Limitations

01 No P&L objective, by design

This model was never validated for trading profitability and none of the checks in this report are a substitute for that. Using its regime read to condition a trading strategy's sizing or risk appetite is a reasonable next step, but that integration and its own validation do not exist yet.

02 Refit stability is moderate, not high

~1 day in 6 over the shared 2009–2014 comparison window gets relabeled as the pipeline sees more data. The automatic state-count selection rule itself is not perfectly stable across vintages either (Section 6) — a full refit today could plausibly land on a different state count than the 4 this report uses.

03 A single, fixed train/holdout split

Both AI stages are fit once on data through 2021-12-31 and frozen — appropriate for validating statistical soundness, but this is not a walk-forward or periodically-retrained system. A live deployment would need a decision about when and how to refit.

04 **Free-daily-data universe**

All 12 tickers come from Yahoo Finance's public chart API — liquid, well-known ETFs and indices, but not the same data quality or point-in-time discipline as a licensed institutional feed. ^IRX (13-week T-bill, originally intended alongside ^TNX for a yield-curve-slope feature) was dropped after its data proved unreliable in repeated fetches.

05 **Regime labels are descriptive, not causal**

The model finds that certain macro/vol/credit conditions co-occur and persist together; it says nothing about what causes a regime shift or how long the current one will last beyond the transition matrix's unconditional base rates.

A **Appendix — Configuration**

Sample window	2007-01-01 →
Train / holdout split	through 2021-
Raw features	16, causal (returns, cross-ratios, realized vo
Autoencoder	16 → 12 → 3 → 12 → 16, ReLU, MSE loss, Adam, lr=1e-3, weight decay=1e-5
HMM	Gaussian emissions, full covariance, 200 E
State-count sweep	2-7, selected via BIC + min. 5-trading-day ave
Selected states	4 — Calm Bull, Crisis, No

Companion interactive dashboard (regime-colored SPY chart with hover detail):
`regime_detection/analysis/regime_dashboard.html` in the project repository.